

Total No. of Questions : 9]

SEAT No. :

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S.E. (Electronics/E & TC) (Electronics & Computer/VLSI Design
& Technology/Electronics & Communication/Advanced

Communication Technology)

ENGINEERING MATHEMATICS - III

(2019 Pattern) (Semester- III) (207005)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Q1 is compulsory.
- 2) Attempt Q2 or Q3, Q4 or Q5, Q6 or Q7, Q8 or Q9.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Figures to the right indicate full marks.
- 5) Use of electronic pocket calculator is allowed.
- 6) Assume suitable data, if necessary.

Q1) Write the correct option for the following multiple choice questions.

a) The second forward difference $\Delta^2 f(x)$ for $f(x) = x^2$, $h = 2$ is given by [2]

- | | |
|--------|--------|
| i) 6 | ii) 12 |
| iii) 4 | iv) 8 |

b) $\nabla(r^2 \log r)$ is equals to [2]

- | | |
|----------------------------|----------------------------|
| i) $r\bar{r}(2\log r + 1)$ | ii) $\bar{r}(2\log r + 1)$ |
| iii) $r(2\log r - 1)$ | iv) $r \log r(r + 1)$ |

c) The residue of $f(z) = \cot z$ at $z = 0$ is [2]

- | | |
|------------|-------|
| i) 1 | ii) 2 |
| iii) π | iv) 0 |

d) The value of $\int_C \bar{F}_0 d\bar{r}$, for $\bar{F} = x\bar{i} + y\bar{j} + z\bar{k}$ along the path $n = t$, $y = t^2$, $z = t^3$ from $t = 0$ to $t = 1$ is [2]

- | | |
|--------------------|-------------------|
| i) $\frac{2}{3}$ | ii) $\frac{3}{2}$ |
| iii) $\frac{5}{3}$ | iv) $\frac{4}{3}$ |

P.T.O.

e) In Runge-Kutta method of fourth order, the value of K_2 is [1]

i) $\frac{h}{2} f(x_0, y_0)$ ii) $hf(x_0, y_0)$

iii) $hf(x_0 + \frac{h}{2}, y_0 + \frac{K_1}{2})$ iv) $f(x_0, y_0)$

f) The poles of $\frac{z^2 + 1}{(z + 1)(z^2 - 9)}$ are [1]

i) $-1, -3, 3$ ii) $1, 3, 4$

iii) $-1, 3, 3$ iv) $1, 2, 3$

Q2) a) Find cubic polynomial by Newton's forward difference formula. Hence find $f(4)$. [5]

x	0	1	2	3
$f(x)$	1	2	1	10

b) Use Simpson's $\left(\frac{1}{3}\right)^{\text{rd}}$ rule to evaluate $\int_0^{\frac{\pi}{2}} \left(\frac{\sin x}{x}\right) dx$ by dividing the interval into 4 equal parts. [5]

c) Using Euler's modified method, find the value of y at $x = 0.05$ given that $\frac{dy}{dx} = x^2 + y, y(0) = 1$ (2 iterations only). [5]

OR

Q3) a) The velocity distribution of fluid near a flat surface is given below [5]

x	0.1	0.3	0.6	0.8
v	0.72	1.81	2.73	3.47

Where x is distance from surface (mm). v is velocity (mm/sec). Use Lagrange's interpolation formula to find v at $x = 0.4$.

b) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ taking $h = \frac{1}{6}$ by using Simpson's $\frac{3^{\text{th}}}{8}$ rule compare with actual value. [5]

c) Use fourth order Runge-Kutta formula to find the value of y when $x = 1$

Given $\frac{dy}{dx} = \frac{y-x}{y+x}$ and $y(0) = 1$ take $h = 1$. [5]

Q4) a) Find the directional derivative of $f(x, y, z) = x^2y + xyz + z^3$ at $(1, 2, -1)$ along normal to the surface $x^2y^3 = 4xy + y^2z$ at the point $(1, 2, 0)$. [5]

b) Show that the vector field $\vec{F} = (y^2 \cos x + z^2)\vec{i} + (2y \sin x)\vec{j} + (2xz)\vec{k}$ is conservative and find scalar field ϕ . Such that $\vec{F} = \nabla \phi$. [5]

c) Show that $\nabla^2 \left[\nabla \cdot \left(\frac{\vec{r}}{r^2} \right) \right] = \frac{2}{r^4}$. [5]

OR

Q5) a) Find the values of a, b, c so that the directional derivative of $\phi = axy^2 + byz + cz^2x^2$ at $(2, 1, 1)$ has a maximum magnitude 12 in the direction parallel to x -axis. [5]

b) If $\vec{F}_1 = yz\vec{i} + zx\vec{j} + xy\vec{k}$, $\vec{F}_2 = (\vec{a} \cdot \vec{r})\vec{a}$ then show that $\vec{F}_1 \times \vec{F}_2$ is solenoidal. [5]

c) For scalar functions ϕ and ψ show that [5]

$$\nabla \cdot (\phi \nabla \psi - \psi \nabla \phi) = \phi \nabla^2 \psi - \psi \nabla^2 \phi$$

Q6) a) Find the work done in moving a particle in a force field $\vec{F} = 3xy\vec{i} - 5z\vec{j} + 10x\vec{k}$ along the curve $x = t^2 + 1, y = 2t^2, z = t^3$ from $t = 1$ to $t = 2$. [5]

b) Using Gauss-Divergence theorem, evaluate $\iiint_V \vec{F} \cdot \hat{n} ds$ where

$\vec{F} = x^3\vec{i} + y^3\vec{j} + z^3\vec{k}$ and s is the surface of the sphere $x^2 + y^2 + z^2 = a^2$. [5]

c) By using Stoke's theorem, evaluate $\iint_s (\nabla \times \vec{F}) \cdot \hat{n} ds$ for

$\vec{F} = y\vec{i} + z\vec{j} + x\vec{k}$ where s is the surface of the paraboloid $z = 1 - x^2 - y^2, z \geq 0$. [5]

OR

Q7) a) A vector field is given by $\vec{F} = \cos y \vec{i} + x(1 - \sin y) \vec{j}$. By using Green's theorem, evaluate $\int_c \vec{F} \cdot d\vec{r}$ where c is the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1, z = 0$. [5]

b) Show that $\iiint_v \frac{dv}{r^2} = \iint_s \frac{\vec{r} \cdot \hat{n}}{r^2} ds$. [5]

c) By using Stoke's theorem, evaluate $\int_c \vec{F} \cdot d\vec{r}$ where $\vec{F} = \sin z \vec{i} + \cos x \vec{j} + \sin y \vec{k}$ and ' c ' is the boundary of rectangle $0 \leq x \leq \pi, 0 \leq y \leq 1$ & $z = 3$. [5]

Q8) a) If $v = 3x^2y - y^3$, then find it's harmonic conjugate u . [5]

b) Evaluate $\oint_c \frac{z^2 + 1}{z - 2} dz$, where c is the circle $|z - 2| = 1$, by using Cauchy's Integral formula. [5]

c) Find the bilinear transformation which maps the points $z = 1, i, 2i$ on the points $w = -2i, 0, 1$ respectively. [5]

OR

Q9) a) If $u = x^4 - 6x^2y^2 + y^4$ then find v such that $f(z) = u + iv$ is analytic & determine $f(z)$ in terms of z . [5]

b) Evaluate $\oint_c \frac{\sin(\pi z^2) + 2z}{(z - 2)} dz$, where c is the circle $|z| = 4$, by Residue theorem. [5]

c) Find the map of the straight line $y = x$ under the transformation $w = \frac{z - 1}{z + 1}$. [5]

